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FIRST PALEOMAGNETIC RESULTS FROM MIO-PLIOCENE SERIES OF THE HELLENIC SEDIMENTARY ARC

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ABSTRACT

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Paleomagnetic measurements have been conducted on the Mio-Pliocene series along the external sedimentary Hellenic arc, in order to detect the presence or absence of rotational deformations in different sections of the arc.

In the central-eastern part of the arc (Crete and Rhodes) paleomagnetic directions are very closely aligned along the North–South direction, showing that this portion of the arc has not undergone significant rotation since at least 7 m.y. (for Crete) and 4 m.y. (for Rhodes).

In the western and northwestern parts, results from South Peloponnesos, Zakinthos, Kephallinia and Corfu indicate that this section of the arc has undergone a clockwise rotation. Detailed analysis of the data from the three Ionian islands shows that their rotation began about 5 m.y. B.P. and has proceeded since then at roughly 5°/m.y., giving a total amplitude of about 26°.

It is suggested that the beginning of this rotation is related to the lower Pliocene compressive phase, which most probably initiated the tectonic activity of the present day Ionian margin.

INTRODUCTION

Among the arcs of the Alpine belt in the Mediterranean area (Fig. 1), the Aegean arc in particular has been investigated in the last 10–15 yrs, from the point of view of geophysics, neotectonics and geodynamical evolution.

Seismic studies of this very active region leave little doubt about the existence of an amphitheatre-shaped slab dipping away from the Hellenic trench area with an average dip of 30–40°, measured from the trench along a radius of the arc. The maximum depth of this slab is 150–200 km, under the Aegean Sea. Moreover, seismic data show a striking difference between the western portion of the Hellenic trench system, which results essentially from underthrusting perpendicular to its

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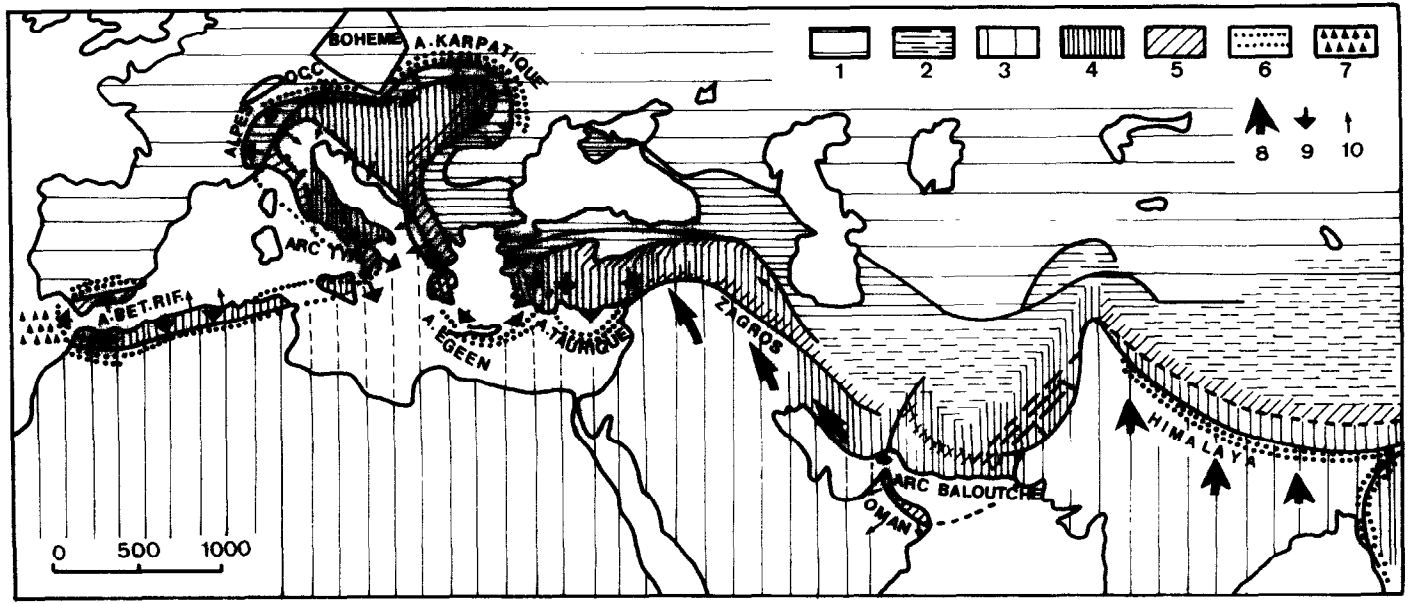


Fig. 1. The arcs of the western Alpine system (from Brunn (1976)). *1* = Euro-Asiatic plate, *2* = margins of this plate involved in the Alpine tectonics, *3* = Indian and Arabian plates, *4* = margins of these plates involved in the Alpine tectonics, *5* = inter-plate domains, *6* = molassic sediments, *7* = olistostromes, *8* = directions of the relative movement of the Indian and Arabic-African plates, the latter being deviated towards the northwest since Eocene times, *9* = direction of the early radial thrusting in the arcs, *10* = direction of the late thrusting towards intra-arc depressions.

direction, and the southeastern portion where the movement is almost parallel to its direction in a zone of sinistral strike-slip faulting. These different motions have given rise to different underwater topographies in the two branches of the trench, which have been studied by seabeam and submarine surveys (Le Pichon et al., 1979a and b).

The external arc along the trenches is subjected to compressional or strike-slip earthquakes whose slip vectors are directed roughly along the N48 direction in the west and along N22 in the east (Le Pichon et al., 1979a and b). The compressional zone of the external arc affects emerged lands only at the northwestern termination, in the Ionian Islands, where these compressional deformations (probably responsible for the present margin) began during the early Pliocene and were still active during the Pleistocene (Mercier et al., 1976). The great extent of these compressive structures makes this region quite different from the rest of Aegea: indeed, the internal Aegean regions have been essentially submitted to a large scale extensional regime since the upper Miocene, and this regime has only occasionally been interrupted by compressive events, often quite moderate, in the upper Miocene–lower Pliocene and lower Pleistocene periods.

Focal mechanism solutions show that the present regime is still extensional in all the regions, even slightly away from the external trenches, such as South Peloponnesos, Crete and Rhodes (McKenzie, 1972, 1978; Ritsema, 1973, 1974). In each of these regions, present-day stress directions, calculated from focal mechanisms or from in-situ measurements, are very close to the stress directions since the middle Pleistocene, as determined from the analysis of neotectonic faults and striations (Paquin et al., 1982, this volume). Thus, neotectonic analysis has allowed us to reconstruct the successive stress patterns in the Aegean arc since the upper Miocene.

Using stress pattern analysis, different authors have considered the problem of the deformation of Aegea, and have suggested models for the geodynamical evolution of the Aegean arc. Even if all the models come to the conclusion that the curvature of the arc has been acquired at least partially because of its tectonic evolution, the geodynamical mechanisms involved are substantially different according to the different authors.

For instance Brunn (1976), Fig. 1, has suggested a model of continuous deformation in which the curvature of the arc is related to the main Tertiary Alpine overthrustings, and was thus essentially completed before the neotectonic period. The extent of thrusting, which is greater in the middle of the arc than at its northwest and southeast terminations (Aubouin, 1973) indeed agrees with the hypothesis of an increase in the curvature of the tectonic pattern during that epoch. This is also related to the maximum abundance of Oligo-Miocene calc-alkaline magmatism in the Aegean Sea behind the central part of the arc (Bellon et al., 1979).

Tapponnier (1977) also concluded that an evolution of the curvature of the arc essentially started in Eocene times as a result of rigid-plastic indentation of Eurasia by peninsulas of the African continent, i.e., the Adriatic peninsula in the west and the Arabic in the east (slip line field theory).

Other models consider that the curvature of the arc has increased substantially during the neotectonic period. McKenzie, for instance, (1970, 1972) had initially suggested that the mechanical interaction of small rigid plates with well-defined margins could suitably describe the main displacements in the Eastern Mediterranean. In a more recent paper (1978) the same author has taken into account the existence of an important deformation of the Aegean arc, as shown by neotectonic studies.

The extent of these internal deformations has led Mercier (1977) and Mercier et al. (1976, 1979) to the notion of the continuum of deformation. In their opinion, the Aegean domain constitutes the plastically deformed edge of the Euro-Asiatic plate. The prolongation of the northern Anatolian strike-slip faults into the northwestern Aegean Sea and the discovery of faults with a sinistral strike-slip component in Macedonia have led Mercier et al. (1979) to consider that the deformation of this region of northern Aegea may be described by a scheme of slip lines, as suggested by Tapponnier. This evolutionary scheme implies rotational deformations of the Aegean lithosphere, clockwise in the northwest and anticlockwise in the southeast of the arc.

Le Pichon and Angelier (1979) consider the neotectonic movement since 13 m.y. to be a first approximation of the clockwise rotation of a rigid plate by approx. 30° about an eulerian pole, located near 40°N, 18°E in the southern Adriatic Sea. An important back arc distension, indicated by intensive normal faulting in the Aegean Sea, is superimposed upon this rigid rotation and modifies its value locally all along the external arc: there is thus maximum rotation in the northwest (Ionian Islands except Corfu) decreasing towards the southeast to attain zero near eastern Crete. According to them, western Crete appears to undergo a clockwise rotation, causing a torsion of the island. More recently (Lybéris et al., 1980, 1981) the discovery of longitudinal variations of deformation in the Kithiran strait as shown by seabeam bathymetric studies and by the existence of en echelon faults, has led these authors to modify their original model. Thus, in the model presented at this conference, Crete is "decoupled" from Peloponnesos and does not show any appreciable rotation.

All these models are based on seismic and tectonic data. An important parameter which cannot be attained by these methods is the rotational deformation and its variations along the arc. Recent progress in both the technology and understanding of the paleomagnetism of sedimentary rocks now allows the use of their weak remanent magnetization for geological purposes. The initially north–south magnetization of the sediments can, in principle, provide a primitive and constant geometrical element along the arc; its deviations with respect to the initial direction in different regions can provide data about the rotational deformations of the arc.

Sedimentary formations of upper Miocene, Pliocene and lower Pleistocene age have thus been sampled along the arc and the paleomagnetic results obtained are reported here.

GEOLOGICAL SETTING AND SAMPLING

The sites were sampled all along the external Aegean arc where the rotational deformations predicted by the available models should be maximal. Their location is shown on the schematic map in Fig. 2.

All the sampled regions have been submitted to quaternary uplift movements which have caused outcropping of large parts of Neogene sedimentary basins. Most of the deposits are of marine origin and thus a stratigraphical age can often be assigned to them, using planktonic Foraminifera or nanofossil stratigraphy.

In the middle and eastern part of the arc, Rhodes, Crete and southern Peloponnesos belong to the "internal domain" of the neotectonic Aegean arc. The sedimentary evolution of these basins has been described in numerous previous works (Drooger and Meulenkamp, 1973; Fortuin, 1978; Guernet, 1978; Angelier, 1979). Essentially, these basins have been submitted to extensional tectonics, sometimes with huge normal faults, during both the sedimentation and the emersion periods. This tectonics was very active since at least the Serravallian (13 m.y.) in Crete (Angelier, 1979) and continued during the deposition of the lacustrine levantine series in Rhodes (Gauthier, 1979) where the deposits are attributed to the upper

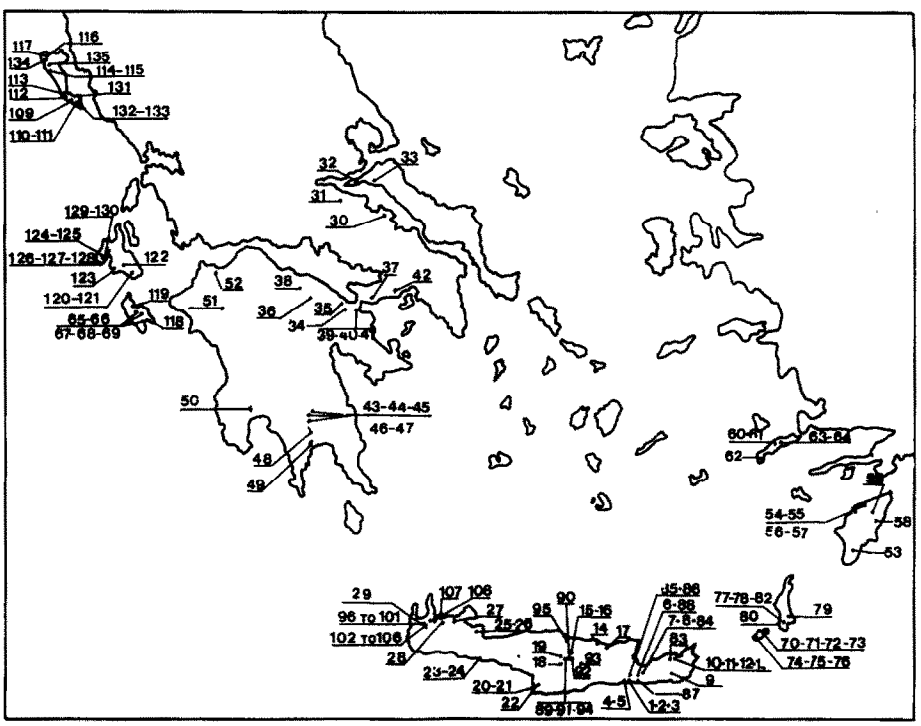


Fig. 2. Schematic map of Aegea showing the position of the sampled sites.

Miocene–middle Pliocene period (Benda et al., 1977; Benda and Meulenkamp, 1979). Some moderate compressional deformations have also occurred after the deposition of Pliocene and Calabrian sediments in Rhodes (Gauthier, 1979), Crete (Angelier, 1979) and in the northern part of the Peloponnesos (Sébrier, 1977). The uplift movement of these regions, sometimes important, was then accompanied by strong, normal faulting since the middle Pleistocene.

In the northwestern part of the arc, Zakinthos, Kephallinia and Corfu belong to the "external domain" of the neotectonic Aegean arc. This zone has essentially been submitted to compressive deformation (Mercier et al., 1976). Four to five km of epicontinental marine carbonates of Mesozoic and Tertiary age characterize the sedimentary sequence of the preapulian isopic zone; this zone constitutes the main part of Kephallinia and Zakinthos and of the submarine continental margin as far as the Adriatic Sea. In this thick carbonate series, the major structures are high-angle reverse faults with NW–SE and NNE–SSW directions, associated with heavy folding (Sorel, 1976; Sorel et al., 1976). The formation of this presently active margin in the northwest most probably started during the strong, lower Pliocene compressive phase (Mercier et al., 1976) which affected all the Ionian Islands and the coastal regions of mainland Greece (Epirus, Akharnania and Elide). (See also Fig. 8.)

The Island of Corfu belongs to the Ionian isopic zone, which was thrust towards the west over the preapulian zone during the lower Pliocene compressive phase. The main structures in the thinner Mesozoic and Tertiary sedimentary sequence of this zone are low-angle overthrusts towards the west or southwest, over a Triassic evaporitic decollement level (IGRS-IFP, 1966); thrusts are associated with dinaric (NW–SE directed) folds and with NE–SW to E–W strike-slip faults, well indicated by dextral movements in Corfu. The neotectonic study of Corfu (Jamet, 1982) shows that this island has undergone the same tectonic evolution as Kephallinia and Zakinthos during the Pliocene and Pleistocene periods.

To conclude this summary of the neotectonic evolution of the northwestern margin of the arc, present data indicate that the submarine margin extends to the northwest farther than Corfu, and that this segment of the margin has undergone the same compressive tectonics which started in lower Pliocene and is still active today.

Sampling sites were chosen on both geological and magnetic criteria. Very fine-grained sediments, clays or marls, sand-free if possible, were preferentially chosen because their sedimentation rate is usually low enough to completely average out the effect of the secular variation of the geomagnetic field over the thickness of a sample. Moreover this kind of material is usually deposited on gentle sedimentary slopes in rather calm waters which is a guarantee of true, distortionless remanent magnetization if magnetic oxides are present.

The colour of the sediments has been carefully observed: indeed many clay sections show a brown-yellow, sometimes rusty colour due to weathering, which could also affect the magnetic oxides with eventual remanent magnetism. Sometimes there is a progressive change of colour towards blue-grey when the surface of the

section is freshened, and sometimes weathering affects the whole section. The blue-grey colour itself is probably not the original colour of the sediments, but a result of chemical reduction phenomena due to their organic contents. Pyritized concretions very frequently observed in Foraminifera shells, for instance, are good evidence of iron migration in the sediment. This transport can take place even after compaction of the sediment, as shown by some macrofossils which have been crushed before being pyritized. Fold tests, however, performed whenever possible, have shown that magnetization has been acquired before the tectonic deformations in all cases. So we have assumed that the reduction phenomena took place at an early stage of the history of the sediments, so that their "magnetic age" is not different from the stratigraphical age.

Care was taken not to sample zones too close to huge faults, and sections containing slump beds or landslide indicators were similarly avoided.

Since the 1979 survey, a portable spinner magnetometer (Chiron et al., 1981) has been used systematically to select sites with a minimum magnetization level, which has been fixed at 10^{-6} emu/cm^3, among those which are geologically significant.

An electrical machine mounted on a frame and powered by a portable generator was used to drill the samples. The corer is a 25-mm bronze barrel, 150 mm long, with a sintered diamond cutting edge. Water was sometimes used as a coolant but, as a routine, compressed air was used to cool the corer and to remove the dust, and this technique proved to be fast and lightweight.

Orientation of cores was done with a special device. Even though the sediments have a very weak magnetism, a sun compass was used whenever possible, because it allows a more precise comparison between the directions of very distant sites (no corrections for regional magnetic anomalies are needed). The accuracy on orientation is about 2°.

A total of more than 2300 cores has been collected from 135 sites in the Aegean arc.

LABORATORY TECHNIQUES AND SITES SELECTION

All the samples were measured prior to any treatment on their arrival at the laboratory with a Digico spinner magnetometer. In general, sites where all the samples had a magnetization intensity lower than $0.8 \cdot 10^{-6}$ emu/cm^3 were discarded at this step, because it was felt that the required accuracy could not be attained in these cases with a spinner magnetometer. Before discarding, however, it was verified by heating at 150–200°C that this low intensity was not due to accidentally conflicting direct viscous and stable reverse magnetization. In some regions it has been impossible to localise formations with this minimum level. This is the case, for instance, for Kephallinia and for some sites in Corfu. Because of the importance of these islands for the understanding of the geodynamical evolution of

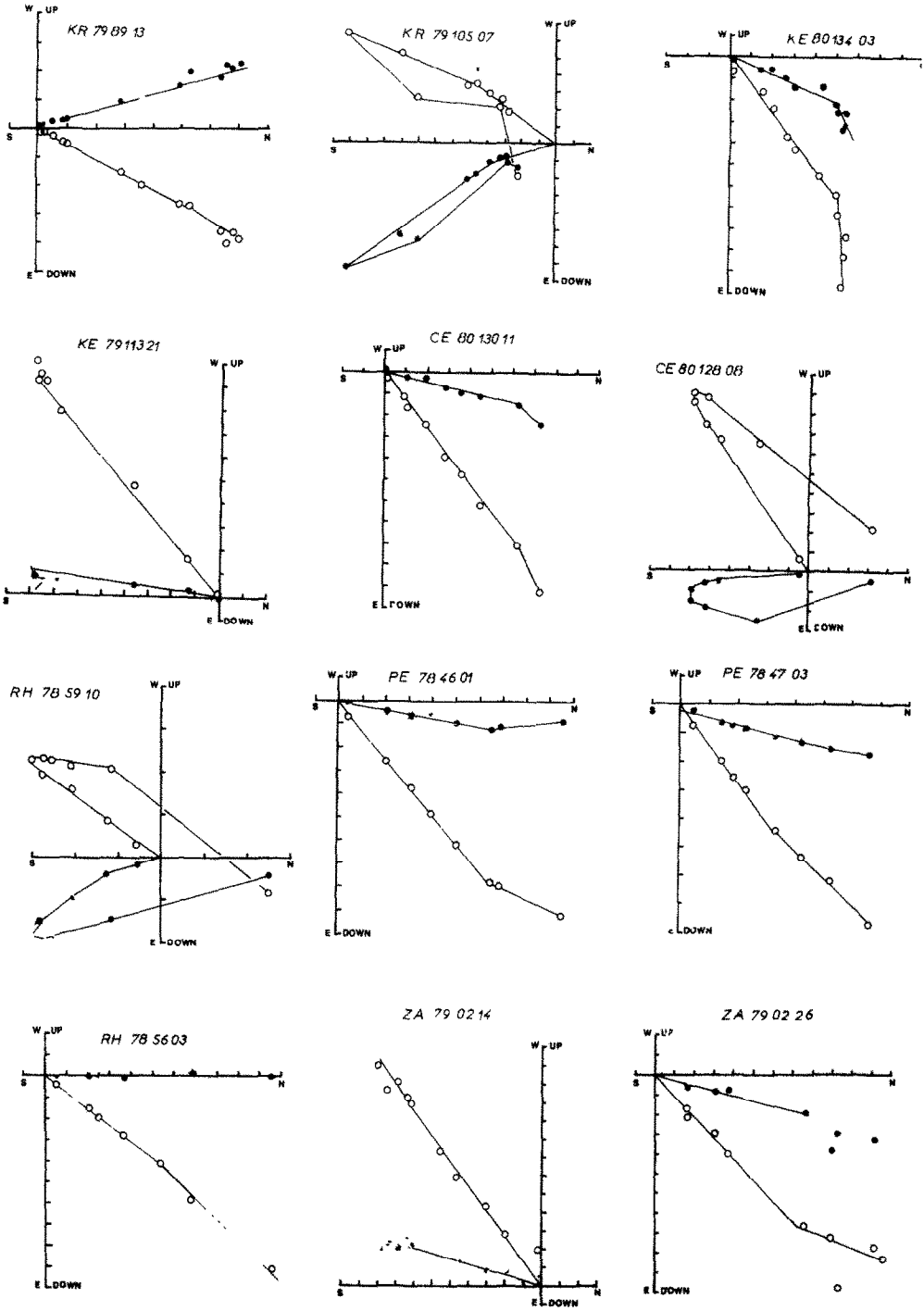


Fig. 3. Thermal demagnetization plots of various samples from different sites. The stable component can be determined with great accuracy.

the arc, sites were nevertheless selected and were studied successfully using the cryogenic magnetometer of the Paleomagnetic Laboratory at Utrecht. Other regions, such as Eubea and Locride (sites 30–33), the Korinthos region of North Peloponnesos (34–42), Kassos (70–76), and some of the sites from Crete were discarded, however, because the magnetization of the collected samples was really too low. Measurements from other regions (Kos, Karpathos) are still in progress.

As we wanted to be sure that no alteration effect was present in our results, we have tried to ascertain the nature of the magnetic oxides. Magnetites are essentially

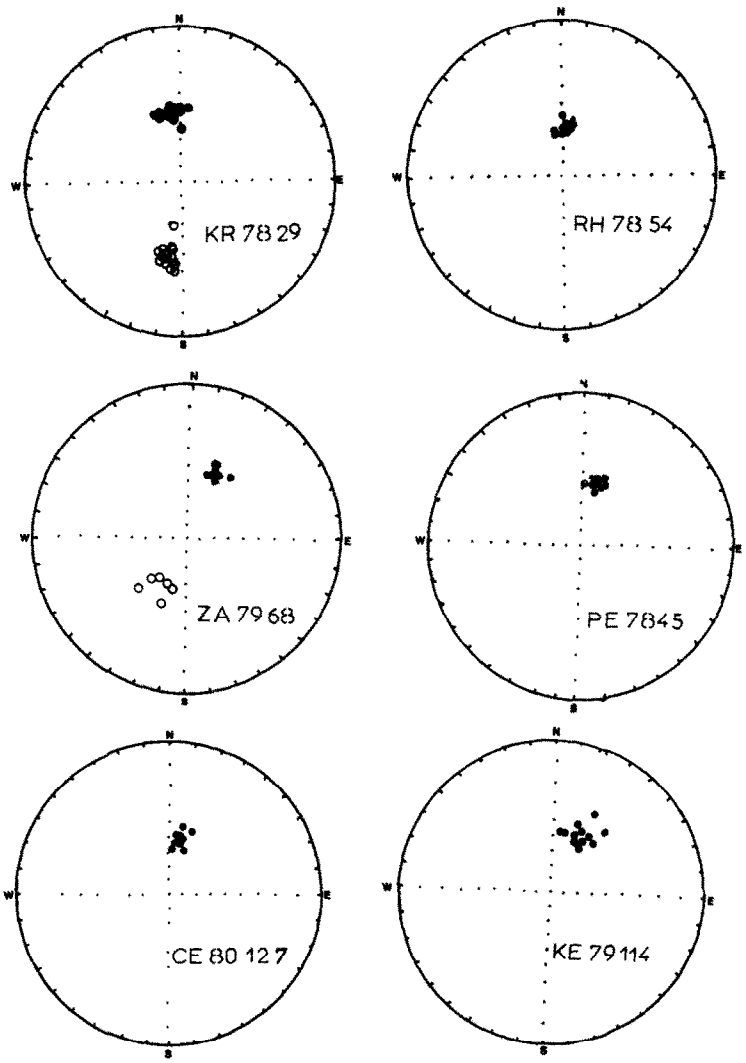


Fig. 4. Stereographic projections showing the intra-site accuracy of the data.

of detritic origin, while haematites may form in situ by different low-temperature oxidation processes, acquiring a CRM whose age may be significantly different from the stratigraphical age of the sediments from the same horizon. So results obtained from haematites-rich sediment must be used with caution, unless it can be proved that magnetization has been acquired at an early stage of the sediment's history.

The distinction between magnetite and haematite-bearing rocks has been made using the IRM acquisition test, applied stepwise up to 12.5 kOe, followed by AC and thermal demagnetization to determine the AC median destructive field of the saturation remanence and the temperature beyond which no magnetization is left. Within the limits of this test, which has been applied on at least two samples from each site, it was found that in 90% of the cases, magnetite was the main magnetic carrier.

The selected sites have then been studied using usual paleomagnetic techniques. AF demagnetization was obtained using a Schönstedt GSD-1 demagnetizer, applying steps of 50 or 100 Oe. Thermal demagnetization was obtained in a laboratory-built furnace having a residual field of about 15 γ. Steps of 50°C were used routinely from 100°C to the highest temperature at which samples could still be measured with sufficient accuracy. This temperature ranges from 400° to 550°C, depending on the samples. Each sample has been step demagnetized and its paleomagnetic direction determined from demagnetization plots. In general, these plots are of good quality and the paleomagnetic directions can thus be determined with reasonable accuracy. Examples of demagnetization plots relating to samples from different regions of the arc are shown in Fig. 3. Figure 4 depicts some stereograms of different sites to give an idea of intra-site scattering.

RESULTS

General

The formations which have been sampled range from the Serravallian–Tortonian boundary (ca. 11.5 m.y.) to the Calabrian stage of the early Quaternary ($\leq$ 1.8 m.y.). A common characteristic of many of these formations is the presence in the samples of a secondary, probably viscous, component of magnetization directed along the present geomagnetic field. Moreover, in most cases the value of the inclination is smaller than the value expected on the basis of a geocentered dipole. This last value is of the order of 55° in Crete and 59° in Corfu. Typical inclination values for the sampled sites are, on the other hand, of the order of 45°, and thus there is a significant inclination error. We believe that this low inclination value is related to the compaction of the sediments, which can be quite important for blue-grey marls, and can be seen in the field by the flattened shape of many macro fossils. This opinion is based on the following observations.

First of all, there is no correlation between the presence or absence of the inclination error and the age or geographical position of the formation. This rules out the possibility that the low value of the inclination is a consequence of an off-centered dipole, because in that case one would reasonably expect a strong correlation with the age of the formation. In the same way, the presence of a local magnetic anomaly would appear as a geographical correlation.

However, the presence of the inclination error seems to be related to the nature of the sediment. For instance, in the calcareous marls of the Kamiros basin in Rhodes, the mean inclination is 51° and so the error is quite small and, in general, seems to be smaller when the amount of carbonate in the sediments is high. It is known that compaction is quite limited in carbonate rich sediments when lithification is rapid. However, an alternative explanation has been pointed out to us by S. Banerjee (pers. commun., 1981): these sediments could also be an example of the well-known phenomenon of "inclination error" exhibited by some laboratory studies of DRM (King, 1955; Griffiths et al., 1960). Both explanations suggest that when the inclination error is observed, the magnetization has been acquired at an early stage of the history of the sediment, and this can be a useful criterion.

Whatever the explanation for their low value, inclination data are certainly not particularly reliable for the sites which we have studied. Declination values, however, are not affected by any of the mechanisms invoked to explain the inclination error; they can thus be safely used as a measure of rotational deformation.

Results from Rhodes

In Rhodes, neogene formations are represented by upper Miocene to middle Pliocene lacustrine series (Levantine) and by the marine transgressive series of upper Pliocene and lower Pleistocene (Calabrian), lying with a major discordance over the pile of the Alpine overthrusts and Oligocene mollasses. Tectonics of these basins is not very important and is mostly synsedimentary (Gauthier, 1979).

Seven sites in total were sampled in the calcareous marls of the Kamiros basin in

TABLE I

Paleomagnetic results from Rhodes

Site	Age	N	D	I	k	α_{95}
RH 7854	Levantine	14	2.6	56	220	2.5
RH 7856	Levantine	14	2.0	42.9	534	1.7
RH 7857	Levantine	6	0.5	52.3	140	4.8
RH 7858	Upper Pliocene	10	2.6	49.3	73.2	5.2
RH 7859	Upper Pliocene	10	177.7	−34.1	39.3	7.0

the north of the island, in the Levantine basin in the south and in the upper Pliocene marine series on the northeast coast.

Two sites were magnetically unstable and all the others gave reliable results, after thermal and AF demagnetization, shown in Table I. The paleomagnetic directions of all these sites are almost exactly aligned with the north–south direction: accordingly, we conclude that Rhodes has not undergone any significant rotation since the lower Pliocene.

Results from Crete

In Crete, Pliocene and upper Miocene (essentially Tortonian) formations have been sampled in the three neogene sedimentary basins of Khania-Castelli in the west, Heraklion-Zaros in the center, and Sitia-Ierapetra in the east of the island. As in Rhodes, the sediments lie with a major discordance over the eroded and faulted pile of the Alpine overthrusts.

Results from sites of Pliocene age

Eleven sites of Pliocene age have been sampled in different regions. Six of them gave reliable results, reported in Table II. From the paleomagnetic direction of each site, the mean paleomagnetic direction for Pliocene formations in Crete can be calculated:

$N = 6, \quad D = 0.1, \quad I = 45.7, \quad k = 62, \quad \alpha_{95} = 7.3.$

This direction lies very close to the north–south axis.

Results from sites of Upper Miocene age

Thirty-six sites of Tortonian age were sampled in Crete. Fifteen of them were found to be sufficiently magnetized and magnetically stable and were studied thoroughly. In many cases, thermal demagnetization gave much more consistent

TABLE II

Paleomagnetic results from sites of Pliocene age in Crete

Site	N	D	I	k	α_{95}
KR 7802	6	360	45	93.5	5.9
KR 7803	9	2.6	36.8	97	4.7
KR 7823	6	175.6	−45.7	22	12
KR 7827	10	0.5	49	184	3.3
KR 7990	10	348	38.6	20	10.1
KR 79108	9	200.5	−57.5	70	5.6

TABLE III

Paleomagnetic results from upper Tortonian sites in Crete. A site contains both direct and reverse directions. The results of direct samples are given first, then those of the reverse ones and finally the overall results (where the reverse samples have been inverted before taking the average)

Region	Site	Polarity	N	D	I	k	α_{95}	VGP	
								Long.	Lat.
East	KR 78-04	N	7	25.1	70	156	4.2	60.3	64.1
	KR 79-83	R	8	157.5	−35	230	3.3	263.3	64.5
	KR 79-84	N	17 } 27	3.7 } 357	37.4 } 37.5	154 } 50	2.7 } 3.8	216.7	75.5
		R	10	169	−37.1	47	6.4		
	KR 79-85	N	8	0	58.6	879	1.7	25.4	85.7
Center	KR 78-17	N	9	4.4	48.1	191	3.4	172.9	82.7
	KR 79-89	N	8	341	37.5	38	7.9	259.7	68.1
	KR 79-91	N	9	358	52	12	13.6	238.4	86.9
	KR 79-93	N	15	11.4	35.5	27	6.4	169.4	71.4
	KR 79-94	N	12	359.4	38.5	23	8.4	207.5	76.4
West	KR 78-29	N	23 } 46	355 } 2.2	42 } 40.5	179 } 99	2.2 } 2.1	193.8	77.5
		R	23	188.6	−38.6	161	2.3		
	KR 79-102	R	9	180.2	−58	109	4.4	25.9	86.8
	KR 79-103	N	5 } 14	345 } 0.7	48 } 48.4	77 } 44	7.1 } 5.9	197.4	84.1
		R	9	191	−49	69	5.6		
	KR 79-105	R	23	165	−44	115	2.7	260.6	73.8
	KR 79-106	N	3 } 10	11.9 } 359.5	43.1 } 41.9	58 } 30	10.6 } 8.0	205.9	78.8
		R	7	175	−36.1	29.6	9.8		
	KR 79-107	R	12	187	−46	118	3.2	165.3	79.9

results than AC demagnetization and thus most of the results reported here have been obtained by thermal demagnetization.

One of the fifteen sites (KR 78-04) gives an anormally high value for the inclination; the fold test suggests that it has been remagnetized after the tectonic tilt, but the magnetic stability is excellent, suggesting that the tilt has been synsedimentary. This hypothesis in turn does not seem reasonable on geological grounds: thus, we have discarded this site. All the others gave very reliable results, reported in Table III. From the paleomagnetic direction of each site, the mean direction for Tortonian formations in Crete can be calculated:

$N = 14, \quad D = 356.5, \quad I = 44.8, \quad k = 52.2, \quad \alpha_{95} = 5.2.$

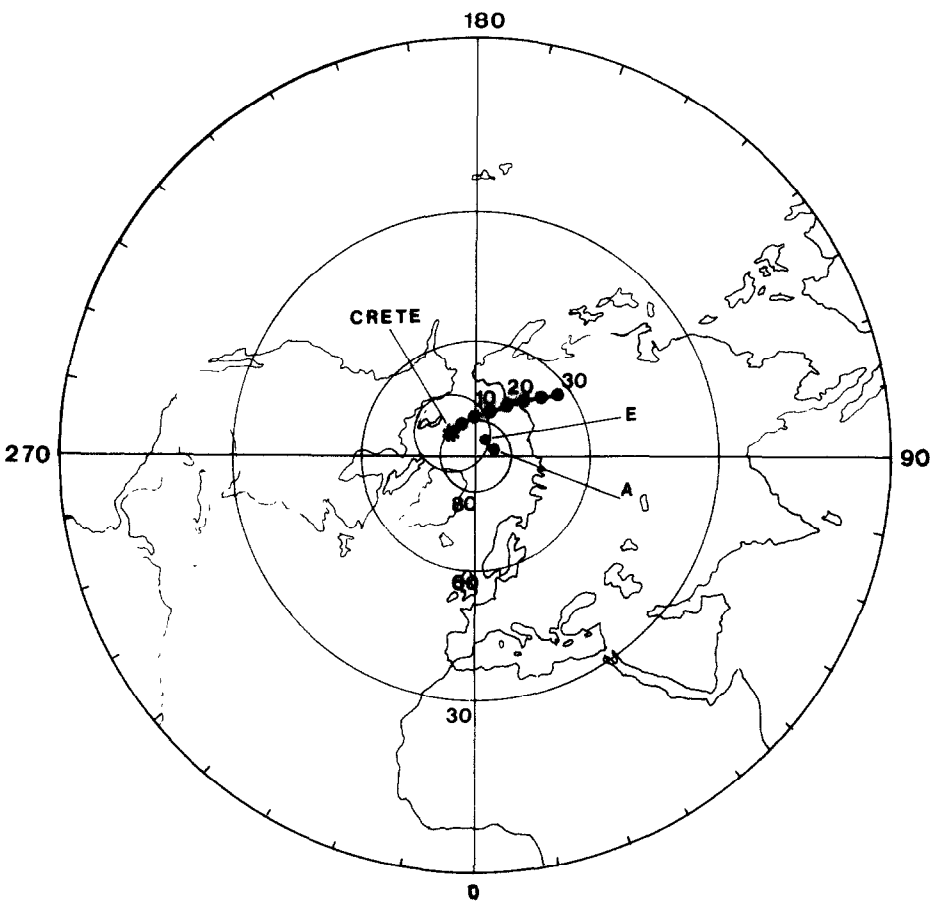


Fig. 5. Stereographic projection showing the position of the calculated paleopole for the Cretan Upper Tortonian formations. The two asterisks are relative to the European (E) and African (A) poles respectively. The full circles are pole position calculated assuming that Crete has undergone clockwise rotation, by steps of 5°, from 0° to 30°. For this calculation, we have used the mean inclination value obtained from Miocene sites (44°8) and not the geocentered dipole value in order to account for sediment compaction.

Alternatively, because the fourteen sites are very much of the same stratigraphical age, one can combine their results to calculate a paleomagnetic pole for Crete. The position of this pole, together with its confidence angle at 95%, is shown in Fig. 5. Also shown are the positions of the European and African poles of the same epoch (McElhinny, 1973; McElhinny and Cowley, 1979) and the position of a hypothetical Cretan pole calculated assuming different values of clockwise rotation for the island. For the inclination in this last case, we have used the mean inclination value measured on the sampled formations and not the geocentric dipole value, which would not have taken account of compaction. It can be seen from Fig. 5 that neither the European nor African poles are significantly different from that obtained for the Cretan Upper Miocene and the data are sufficiently precise that a rotation as small as 10° or 15° should be detected. We conclude that Crete has not undergone a significant rotation with respect to Europe or Africa since Tortonian times.

A more detailed description of the results obtained in Crete is given elsewhere (Valente et al., 1982; Valente, 1982).

Results from southern Peloponnesos

In the southern part of Peloponnesos, the gulfs of Messenia and Laconia are extended inland by two grabens separated by the horst of the Taygetos mountains. The sites were sampled in the sediments which were deposited in these depressions.

Site 7850, near Messenia, has been sampled in shallow water marine sediments of Calabrian age. In Laconia, site 7849, near Krokeai, was sampled in marine-brackish sediments and sites 7845, 7846, 7847, near Skoura, in lacustrine sediments which are considered to be lateral continental equivalents of the Laconian and Messenian gulfs

TABLE IV

Paleomagnetic results from southern Peloponnesos

Site	Age	N	D	I	k	α_{95}
PE 7845	Middle Pliocene to Calabrian	14	18.6	46.0	433	1.8
PE 7846	Middle Pliocene to Calabrian	9	15.3	54.2	587	2.4
PE 7847	Middle Pliocene to Calabrian	11	18.5	53.7	438	2.0
PE 7849	Middle Pliocene to Calabrian	9	194.3	−36.8	84.3	5.1
PE 7850	Calabrian	10	177.7	−44.3	157	3.5

(Dufaure, 1975). Thus the age of these lacustrine deposits is between ca. 3 and 1.5 m.y.

The paleomagnetic results obtained from these sites are shown in Table IV. It can be seen that all but the Calabrian site show a marked deviation from the north–south direction which could be interpreted as a clockwise rotation.

Results from the Ionian islands

In the northwestern region of the Aegean Arc, sites were sampled in both the Preapulian and Ionian zones.

In Zakinthos, sites were selected in blue-grey marls of Serravallian–Tortonian boundary age (Bizon, 1967; Benda and Meulenkamp, 1979) in the pre-Apulian zone and from Pliocene and Calabrian sections of the Ionian zone in the south of the island.

In Kephallinia, all the results were obtained from sites located in the Lyxourion peninsula which belongs to the pre-Apulian zone.

Corfu is entirely in the Ionian zone. Sites were selected in the four main neogene basins: the Karousadhes basin in the north, the Giannadhes and the Potamos monoclines in the center and the Lefkimi monocline in the south.

Seven sites in total were chosen in Zakinthos, but only three of them were found to be magnetically stable.

The results are shown in Table V. It can be seen that the two sites from the Serravallian–Tortonian boundary show a significant deviation from the north–south direction.

The results obtained from Kephallinia are shown in Table VI. Four sites gave very reliable results; all the others were found to be unstable or too weakly magnetized.

A total of thirteen sites were selected in Corfu, some of them (79113, 79114, 80134) covering a geographical distance of $\simeq 2$ km. Table VII gives the results obtained: nine sites were very stable and reliable, three were unstable and one has

TABLE V

Paleomagnetic results from Zakinthos

Site	Age	N	D	I	k	α_{95}
ZA 80 118	Calabrian NN 19	10	184	−57.0	84.5	4.8
ZA 80 119	Serravalian	7	206.8	−39.8	248	3.5
ZA 79 66	Serr-Torton boundary NN 9	15	25.3	44.9	76.6	4.1

TABLE VI

Paleomagnetic results from Kephallinia

Site	Age	N	D	I	k	α_{95}
CE 80 127	Upper Pliocene NN 18 (D. Brouwery) G. inflata	11	12.2	50.2	186	3.1
CE 80 129	Upper Pliocene NN 18 (D. Brouwery) G. inflata	14	11.9	52.2	164.4	2.9
CE 80 130	Upper Pliocene NN 18 (D. Brouwery) G. inflata	9	11.7	55.9	364	2.5
CE 80 128	Calabrian NN 19	11	179	−53.7	337	2.3

TABLE VII

Paleomagnetic results from Corfu

Site	Age	N	D	I	k	α_{95}
KE 79 114	Tortonian G. acostaensis	11	29.2	43.8	65.4	5.2
KE 80 135	Messinian	11	205.2	−47.5	61.8	5.3
KE 79 113	Lower Pliocene NN 12	12	15.8	53.4	131	3.5
KE 80 134	Lower Pliocene G. margaritae	32	19.4	41.9	46.6	3.6
KE 79 117	Lower Pliocene NN 14	8	202.6	−20.6	30.9	7.9
KE 79 115	Lower Pliocene G. margaritae	10	26.3	32	50.6	6.2
KE 79 111	Middle Pliocene G. puncticulata NN 15	15	14	57.2	160	2.8
KE 79 116	Middle Pliocene G. crassaformis	5	2.3	55	558	2.6
KE 80 133	Middle Pliocene G. crassaformis	8	340	42.3	129	4.3

been eliminated because the intrasite dispersion was too great to yield a reliable paleomagnetic direction ($K < 15$). It can be seen from this table that one site (KE80133) indicates an anticlockwise rotation of 20°, while all the others indicate clockwise rotations of variable amplitude according to the age of the formations. Landslide indicators have been discovered in the close proximity of this site, though not on the site itself. We think, however, that a landslide is the most plausible explanation for the results from this site, which is the only one among all the sites of the western portion of the arc to exhibit anticlockwise rotation.

DISCUSSION

The first conclusion which can be drawn from the results from the 47 sites reported above is that the eastern and southeastern portions of the arc have not undergone any significant rotation since Upper-Tortonian times, while the western and northwestern sections show a very definite clockwise rotation. This situation is schematically depicted in Fig. 6 where the two symbols, curled arrow and circle,

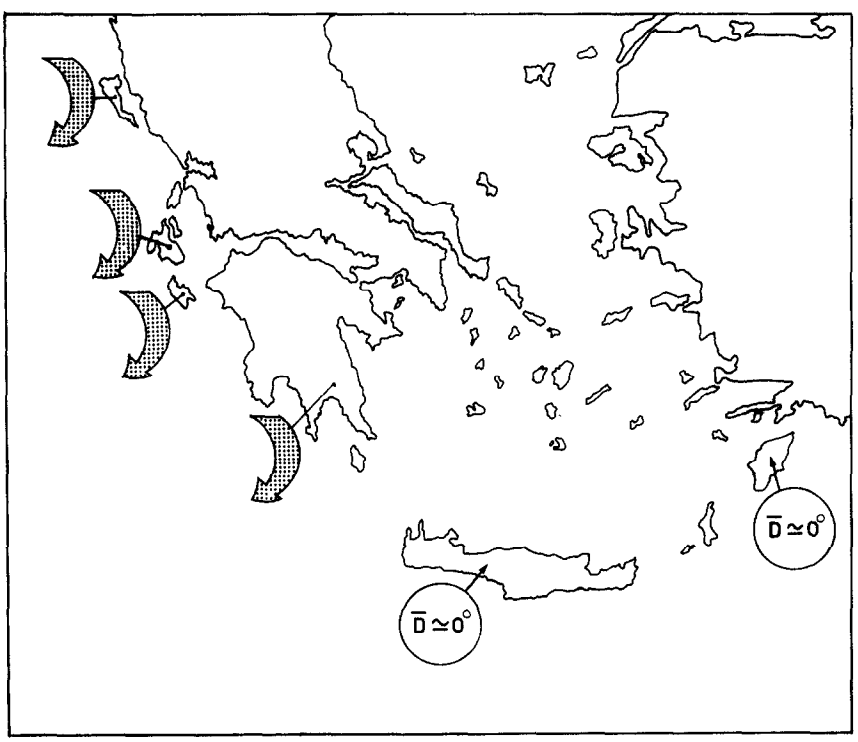


Fig. 6. Schematic map of Aegea showing qualitatively that the western and northwestern sections are submitted to clockwise rotations (curled arrows) while no rotations have been detected in either Crete or Rhodes (circles). The transition between these two regimes is between Crete and southern Peloponnesos.

indicate clockwise rotation and absence of rotation, respectively. Paleomagnetic measurements cannot detect small scale translation movements, so nothing else can be inferred from our data for Crete and Rhodes.

It thus appears necessary to postulate the existence of some sort of geological mechanism "decoupling" Crete from Peloponnesos. There is very good agreement, in this respect, between these data, the geological observations of Lybéris et al. (1981) and the model suggested by Angelier et al. (1982). This last model is also in agreement with the overall picture given by paleomagnetic data, i.e., a non-rotating southeastern section and a rotating northwestern one, with the only difference being that the model does not extend as far north as Corfu, but only to Kephallinia.

Leaving aside results from Peloponnesos which can only be used qualitatively because the age of the formations is ill defined, one can obtain quantitative information about the rotation by superposing the results from the Ionian Islands on a single diagram relating the measured angle of rotation to the age of the corresponding formation. This diagram is shown in Fig. 7 (the single site from Corfu giving anticlockwise rotation has been omitted from this plot).

In this diagram the broken line superposed to the data has been obtained by usual linear regression of the data performed separately in the interval 0–5.5 m.y. (eleven points) and in the interval 5.5–11.5 m.y. (four points). In this last interval the calculated line is very closely horizontal, as shown in the diagram. The shaded area represents a rough estimate of the error which has been calculated, taking into account only the errors on the paleomagnetic data and not on the age of the formations. The real uncertainty is thus definitely greater.

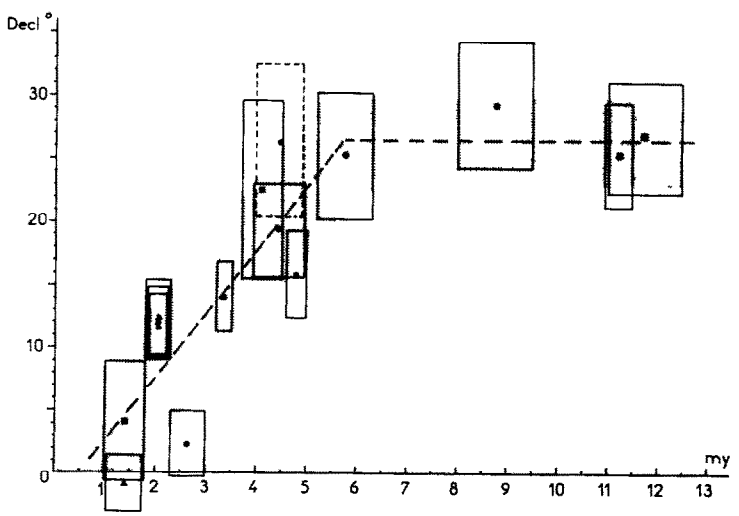


Fig. 7. Plot of the measured angle of rotation versus the age of the corresponding formations for the Ionian Islands: closed square—Zackinthos, closed triangle—Kephallinia, closed circle—Corfu.

Within the limits of this accuracy, it can be seen from this diagram that data from the three Ionian Islands all fall on the same curve, which is a consequence and at the same time an additional proof of their structural unity.

It can also be seen from the diagram that the angle of rotation increases with the age of the corresponding formations up to about 5 m.y. and then remains roughly constant up to 11.5 m.y. This diagram is thus good evidence that the rotational

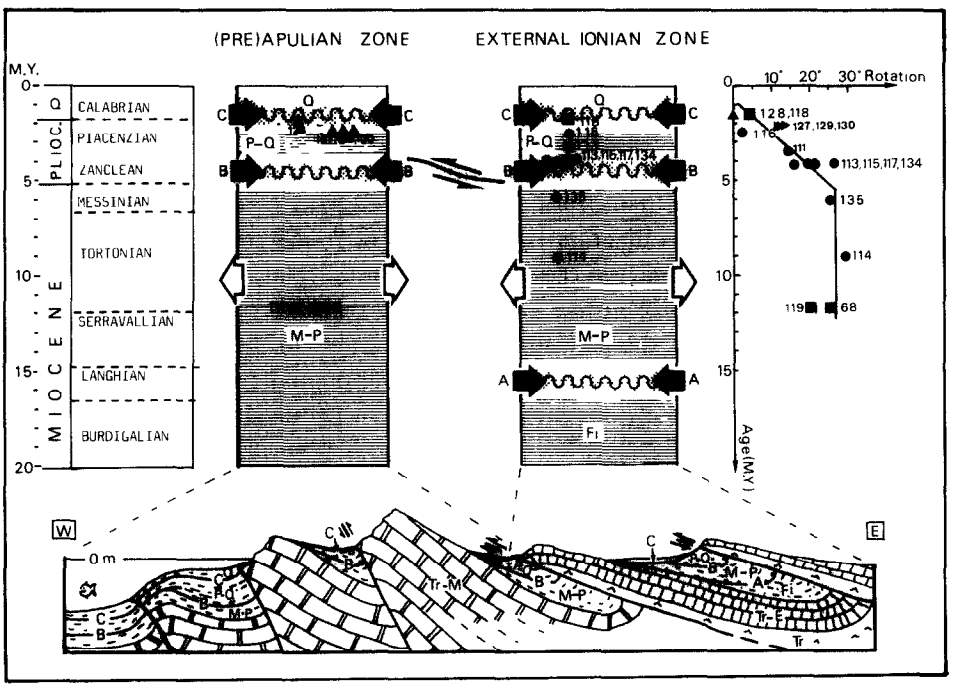


Fig. 8. Geological setting of the sampling in the Ionian Islands: summary of the tectonic and sedimentary evolution.

Q = continental and marine middle Pliocene to Present. Pre-Apulian Zone: M-P = concordant marls middle Miocene to lower Pliocene; P-Q = Pliocene to Calabrian discordant series.

The discordances (A, B, C) are relative to the main compressive phases: A = middle Miocene in the Ionian zone only; B = lower Pliocene in both zones. Present active margin was probably initiated by this phase, which in addition reactivated folds and thrusts in the Ionian zone; C = lower Quaternary (post-Calabrian) compressive phase affecting both zones and followed by persistent compressive episodes until now.

Four sites (68 and 119 in Zakinthos, 114 and 135 in Corfu) belong to a period of synsedimentary distension preceding the lower Pliocene compressive phase. Paleomagnetic results concerning the rotation are reported on the right on the same time scale for comparison with the tectonic evolution.

The schematic cross-section (length $\simeq$ 40 km, height $\simeq$ 5 km) shows the tectonic styles of the pre-Apulian zone (including the submarine structures of the margin) and of the external Ionian zone. Symbols are as follows: Ionian zone: Tr = Triassic evaporites; Tr-E = Triassic–Eocene carbonate series; Fl = flysch, ending in Burdigalian; $M.P.$ = transgressive marine molasse, middle Miocene to lower Pliocene; $P.Q.$ = Pliocene to lower Pleistocene (Calabrian), second transgressive marine series.

movement of the Ionian islands was zero between 11.5 and about 5 m.y. B.P. and has proceeded since then at a mean rate of about 5° per m.y.

Because of the lack of data, especially in the 11.5–5 m.y. interval, no precise age can be assigned to the beginning of the rotation. This can only be estimated at 5 ± 1 m.y. The results are sufficiently precise, however, to reject the hypothesis of a rotation starting at 13 m.y. B.P., which has been proposed by Le Pichon and Angelier (1979). The maximum angle of rotation, of the order of 26°, is, on the other hand, in excellent agreement with the predictions of this model.

We are well aware that our results need to be improved in number and accuracy before any definitive interpretation can be given. Nevertheless, we would like to show that the present data can be interpreted within the framework of the known neotectonic evolution of this area. This is done schematically in Fig. 8 which shows this evolution for both the pre-Apulian and Ionian zones, the stratigraphical position of the sampled sites in each zone and, on the right-hand side, the corresponding measured rotations.

It can be seen that the "knee" of the angle-of-rotation-versus-age plot overlaps the lower Pliocene compressive phase in age. During the upper Miocene, the two zones are submitted to an extensional regime and, according to Mercier et al. (1976), the compressive phase marks the moment when the external Aegean became an active continental margin under compression. We suggest that the beginning of the rotation of the Ionian Islands is related to the formation of this margin.

Our present data are certainly not accurate enough to prove this suggestion, i.e., that the beginning of the rotation and the lower Pliocene compressive phase are *exactly* simultaneous. However, it seems unreasonable to us that a rotation could be initiated in an extensive regime and continue unaffected when this regime changes to a compressive one in such a striking and sudden way. Future work will thus be devoted to improve the accuracy of the angle-of-rotation-versus-age plot.

To conclude, we have shown that while the eastern portion of the arc has not undergone significant rotation, the northwestern part has rotated clockwise at an angle of about 26°, and the beginning of this rotation can be tentatively correlated with the lower Pliocene compressive phase.

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